Math 199, Spring 2026
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Week 2 - Propositions

Goals: Understand how to work with propositions and the various relations between propositions. Develop a unified understanding of the basic building blocks of "mathematical" logic in both formal and colloquial settings. As always, practice getting better at proving things.

1) In this problem, we will find a so-called *Ramsey number*, $R(3,3) = 6$. Call a pair of people **friends** if they have met each other, and **strangers** if they have not.

(a) Show that there can exist a group of five people within which there is neither three mutual friends nor three mutual strangers. First, try to reformulate this as a precise mathematical statement. *Hint:* dots and colored edges can be helpful.

(b) Show that in any group of six people, there must exist within them either three mutual friends or three mutual strangers.

(c) Given positive integers n, m, what do you think the number $R(n, m)$ represents? Fun fact: $R(5, 5)$ is still not known!

2) Write in words what each of the following symbols means. Give one example using each symbol in a grammatical english sentence as if the symbol were a word. The sentence can be about math or not.

(a) $\forall$

(b) $\exists$

(c) $\implies$

(d) $\iff$

(e) $\neg$

(e) $\wedge$

(f) $\vee$

3) Let P, Q, R, and S be the following statements (a.k.a. propositions). (*Warning:* This problem will have some impercise statements and tasks. It is a good habit to recognize when this is the case, and it is also a good mathematical skill to be able to reason about things like this anyway!)

P : *It snowed in Chambana recently.*

Q : *The ground in Chambana will be wet at some point in the next six months.*

R : *The ground in Chambana will have snow on it for the next month.*

S : *The ground in Chambana will have ice on it for the next month.*

(a) Give a realistic counterexample to show that the statement $P \implies Q$ might not be true. Comment on whether or not you think that the statement $P \implies Q$ has been true for the past 100 years.

For (b)-(d) we will assume that $P \implies Q$ is true. (*Bonus:* Explain how we can edit these parts of the problem to both not make this assumption and not change the content of the problem at all.)

(b) Is the statement $Q \implies P$ necessarily true? Explain why or why not.

(c) Is the statement $Q \implies P$ necessarily false? Explain why or why not.

(d) Is the statement $Q \implies P$ probably true? Either explain why it is or give a likely counterexample.

(e) Is the statement $P \implies (R \vee S)$ probably true? Either explain why it is or give a likely counterexample.

(f) Give a realistic explanation for how the statement $P \implies (R \vee S)$ might be true while the statement $(P \implies R) \wedge (P \implies S)$ is false.

(g) Give a realistic explanation for how the statement $P \implies (R \vee S)$ might be true while the statement $(P \implies R) \wedge (P \implies S)$ is false. Formulate the claim of this problem as a proposition in symbols.

(h) Is the statement $[(P \implies R) \wedge (P \implies S)] \implies [P \implies (R \vee S)]$ necessarily true? Explain why or why not.

(i) Is the statement $(\neg[P \implies (R \vee S)]) \implies (P \implies Q)$ probably true? Give a "proof" of your answer.

(j) Given that all the usual laws of nature as you know them are true, is the statement in (i) *necessarily* true? Give a proof or counterexample.

(k) Describe one or two ways of modifying this problem to make all combinations of the statements P, Q, R, S precise statements that can (in theory) be proven true or false.